

```
In [1]: import pandas as pd
```

```
In [3]: # Cargar la base de datos de iris
iris1 = pd.read_csv("iris.csv")

iris1
```

```
Out[3]:
```

	sepal.length	sepal.width	petal.length	petal.width	variety
0	5.1	3.5	1.4	0.2	Setosa
1	4.9	3.0	1.4	0.2	Setosa
2	4.7	3.2	1.3	0.2	Setosa
3	4.6	3.1	1.5	0.2	Setosa
4	5.0	3.6	1.4	0.2	Setosa
...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	Virginica
146	6.3	2.5	5.0	1.9	Virginica
147	6.5	3.0	5.2	2.0	Virginica
148	6.2	3.4	5.4	2.3	Virginica
149	5.9	3.0	5.1	1.8	Virginica

150 rows × 5 columns

```
In [5]: etiquetas = iris1["variety"]
etiquetas
```

```
Out[5]: 0      Setosa
1      Setosa
2      Setosa
3      Setosa
4      Setosa
...
145    Virginica
146    Virginica
147    Virginica
148    Virginica
149    Virginica
Name: variety, Length: 150, dtype: object
```

```
In [7]: from numpy import array
from sklearn.decomposition import TruncatedSVD

A = iris1
# Remover columna no numérica
```

```
A = A.drop(["variety"], axis=1)
A
```

```
Out[7]:
```

	sepal.length	sepal.width	petal.length	petal.width
0	5.1	3.5	1.4	0.2
1	4.9	3.0	1.4	0.2
2	4.7	3.2	1.3	0.2
3	4.6	3.1	1.5	0.2
4	5.0	3.6	1.4	0.2
...	...	...	...	...
145	6.7	3.0	5.2	2.3
146	6.3	2.5	5.0	1.9
147	6.5	3.0	5.2	2.0
148	6.2	3.4	5.4	2.3
149	5.9	3.0	5.1	1.8

150 rows × 4 columns

```
In [9]: from numpy import diag
from numpy import zeros
from scipy.linalg import svd
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

```
In [11]: U, s, VT = svd(A) # Obtención de matrices cuyo producto arroja la matriz A
Sigma = zeros((A.shape[0], A.shape[1])) # Creación de matriz de ceros de la misma
Sigma[:A.shape[1], :A.shape[1]] = diag(s)
```

```
In [13]: print("Matriz U: ")
print(U)
```

```
Matriz U:
[[-0.06161685  0.12961144  0.0021386  ... -0.09343429 -0.09573864
  -0.08085465]
 [-0.05807094  0.11101978  0.07067239 ...  0.03690405 -0.03153954
  0.01309526]
 [-0.05676305  0.11796647  0.00434255 ...  0.03066199  0.19531473
  0.13569909]
 ...
 [-0.0940593  -0.0498297  -0.04144001 ...  0.98181631 -0.02194514
  -0.00894446]
 [-0.09488961 -0.05610123 -0.21297821 ... -0.02155617  0.94178018
  -0.02971961]
 [-0.08847836 -0.0515697  -0.09575285 ... -0.0086052  -0.03021088
  0.9736599  ]]
```

```
In [15]: print("Matriz Sigma: ")  
         print(Sigma)
```

[illegible]

[illegible]

[illegible]

```
In [17]: print("Matriz VT: ")
          print(VT)
```

Matriz VT:

```
[[-0.75110816 -0.38008617 -0.51300886 -0.16790754]
 [ 0.2841749   0.5467445  -0.70866455 -0.34367081]
 [ 0.50215472 -0.67524332 -0.05916621 -0.53701625]
 [ 0.32081425 -0.31725607 -0.48074507  0.75187165]]
```

```
In [19]: # Para reconstruir la matriz original
B = U.dot(Sigma.dot(VT))
print(B)
```

[[5.1 3.5 1.4 0.2]
[4.9 3. 1.4 0.2]
[4.7 3.2 1.3 0.2]
[4.6 3.1 1.5 0.2]
[5. 3.6 1.4 0.2]
[5.4 3.9 1.7 0.4]
[4.6 3.4 1.4 0.3]
[5. 3.4 1.5 0.2]
[4.4 2.9 1.4 0.2]
[4.9 3.1 1.5 0.1]
[5.4 3.7 1.5 0.2]
[4.8 3.4 1.6 0.2]
[4.8 3. 1.4 0.1]
[4.3 3. 1.1 0.1]
[5.8 4. 1.2 0.2]
[5.7 4.4 1.5 0.4]
[5.4 3.9 1.3 0.4]
[5.1 3.5 1.4 0.3]
[5.7 3.8 1.7 0.3]
[5.1 3.8 1.5 0.3]
[5.4 3.4 1.7 0.2]
[5.1 3.7 1.5 0.4]
[4.6 3.6 1. 0.2]
[5.1 3.3 1.7 0.5]
[4.8 3.4 1.9 0.2]
[5. 3. 1.6 0.2]
[5. 3.4 1.6 0.4]
[5.2 3.5 1.5 0.2]
[5.2 3.4 1.4 0.2]
[4.7 3.2 1.6 0.2]
[4.8 3.1 1.6 0.2]
[5.4 3.4 1.5 0.4]
[5.2 4.1 1.5 0.1]
[5.5 4.2 1.4 0.2]
[4.9 3.1 1.5 0.2]
[5. 3.2 1.2 0.2]
[5.5 3.5 1.3 0.2]
[4.9 3.6 1.4 0.1]
[4.4 3. 1.3 0.2]
[5.1 3.4 1.5 0.2]
[5. 3.5 1.3 0.3]
[4.5 2.3 1.3 0.3]
[4.4 3.2 1.3 0.2]
[5. 3.5 1.6 0.6]
[5.1 3.8 1.9 0.4]
[4.8 3. 1.4 0.3]
[5.1 3.8 1.6 0.2]
[4.6 3.2 1.4 0.2]
[5.3 3.7 1.5 0.2]
[5. 3.3 1.4 0.2]
[7. 3.2 4.7 1.4]
[6.4 3.2 4.5 1.5]
[6.9 3.1 4.9 1.5]
[5.5 2.3 4. 1.3]
[6.5 2.8 4.6 1.5]
[5.7 2.8 4.5 1.3]

[6.3 3.3 4.7 1.6]
[4.9 2.4 3.3 1.]
[6.6 2.9 4.6 1.3]
[5.2 2.7 3.9 1.4]
[5. 2. 3.5 1.]
[5.9 3. 4.2 1.5]
[6. 2.2 4. 1.]
[6.1 2.9 4.7 1.4]
[5.6 2.9 3.6 1.3]
[6.7 3.1 4.4 1.4]
[5.6 3. 4.5 1.5]
[5.8 2.7 4.1 1.]
[6.2 2.2 4.5 1.5]
[5.6 2.5 3.9 1.1]
[5.9 3.2 4.8 1.8]
[6.1 2.8 4. 1.3]
[6.3 2.5 4.9 1.5]
[6.1 2.8 4.7 1.2]
[6.4 2.9 4.3 1.3]
[6.6 3. 4.4 1.4]
[6.8 2.8 4.8 1.4]
[6.7 3. 5. 1.7]
[6. 2.9 4.5 1.5]
[5.7 2.6 3.5 1.]
[5.5 2.4 3.8 1.1]
[5.5 2.4 3.7 1.]
[5.8 2.7 3.9 1.2]
[6. 2.7 5.1 1.6]
[5.4 3. 4.5 1.5]
[6. 3.4 4.5 1.6]
[6.7 3.1 4.7 1.5]
[6.3 2.3 4.4 1.3]
[5.6 3. 4.1 1.3]
[5.5 2.5 4. 1.3]
[5.5 2.6 4.4 1.2]
[6.1 3. 4.6 1.4]
[5.8 2.6 4. 1.2]
[5. 2.3 3.3 1.]
[5.6 2.7 4.2 1.3]
[5.7 3. 4.2 1.2]
[5.7 2.9 4.2 1.3]
[6.2 2.9 4.3 1.3]
[5.1 2.5 3. 1.1]
[5.7 2.8 4.1 1.3]
[6.3 3.3 6. 2.5]
[5.8 2.7 5.1 1.9]
[7.1 3. 5.9 2.1]
[6.3 2.9 5.6 1.8]
[6.5 3. 5.8 2.2]
[7.6 3. 6.6 2.1]
[4.9 2.5 4.5 1.7]
[7.3 2.9 6.3 1.8]
[6.7 2.5 5.8 1.8]
[7.2 3.6 6.1 2.5]
[6.5 3.2 5.1 2.]
[6.4 2.7 5.3 1.9]

```

[6.8 3.  5.5 2.1]
[5.7 2.5 5.  2. ]
[5.8 2.8 5.1 2.4]
[6.4 3.2 5.3 2.3]
[6.5 3.  5.5 1.8]
[7.7 3.8 6.7 2.2]
[7.7 2.6 6.9 2.3]
[6.  2.2 5.  1.5]
[6.9 3.2 5.7 2.3]
[5.6 2.8 4.9 2. ]
[7.7 2.8 6.7 2. ]
[6.3 2.7 4.9 1.8]
[6.7 3.3 5.7 2.1]
[7.2 3.2 6.  1.8]
[6.2 2.8 4.8 1.8]
[6.1 3.  4.9 1.8]
[6.4 2.8 5.6 2.1]
[7.2 3.  5.8 1.6]
[7.4 2.8 6.1 1.9]
[7.9 3.8 6.4 2. ]
[6.4 2.8 5.6 2.2]
[6.3 2.8 5.1 1.5]
[6.1 2.6 5.6 1.4]
[7.7 3.  6.1 2.3]
[6.3 3.4 5.6 2.4]
[6.4 3.1 5.5 1.8]
[6.  3.  4.8 1.8]
[6.9 3.1 5.4 2.1]
[6.7 3.1 5.6 2.4]
[6.9 3.1 5.1 2.3]
[5.8 2.7 5.1 1.9]
[6.8 3.2 5.9 2.3]
[6.7 3.3 5.7 2.5]
[6.7 3.  5.2 2.3]
[6.3 2.5 5.  1.9]
[6.5 3.  5.2 2. ]
[6.2 3.4 5.4 2.3]
[5.9 3.  5.1 1.8]]

```

```

In [21]: # Reducción a DOS Dimensiones
n_elements = 2

UReduced2 = U[:, :n_elements]
print("U Reducida:")
print(UReduced2)

```

U Reducida:

[[-6.16168450e-02	1.29611444e-01]
[-5.80709402e-02	1.11019776e-01]
[-5.67630474e-02	1.17966465e-01]
[-5.66534426e-02	1.05308145e-01]
[-6.12302023e-02	1.31089790e-01]
[-6.75031684e-02	1.30884835e-01]
[-5.74820774e-02	1.16598182e-01]
[-6.09726328e-02	1.20943120e-01]
[-5.37611959e-02	9.99414853e-02]
[-5.88266594e-02	1.12043088e-01]
[-6.52918230e-02	1.36578094e-01]
[-5.99417779e-02	1.13753142e-01]
[-5.71132323e-02	1.11354756e-01]
[-5.15957540e-02	1.15324791e-01]
[-6.80071908e-02	1.64183057e-01]
[-7.07625894e-02	1.59056485e-01]
[-6.53647385e-02	1.46844820e-01]
[-6.17918218e-02	1.27676473e-01]
[-6.92802967e-02	1.34541441e-01]
[-6.35146947e-02	1.32921488e-01]
[-6.51727724e-02	1.19363091e-01]
[-6.32935829e-02	1.27908181e-01]
[-5.59608478e-02	1.40649812e-01]
[-6.29534207e-02	1.05679869e-01]
[-6.15456003e-02	1.01783154e-01]
[-5.99228863e-02	1.04639775e-01]
[-6.18571938e-02	1.13083182e-01]
[-6.29341837e-02	1.27221439e-01]
[-6.20034877e-02	1.28133097e-01]
[-5.83668698e-02	1.05996477e-01]
[-5.87535125e-02	1.04518131e-01]
[-6.44535110e-02	1.23473141e-01]
[-6.51357378e-02	1.47626432e-01]
[-6.75203891e-02	1.57559767e-01]
[-5.90016362e-02	1.10108117e-01]
[-5.85766335e-02	1.26756434e-01]
[-6.42131623e-02	1.40001403e-01]
[-6.02724944e-02	1.31424770e-01]
[-5.36226769e-02	1.07009819e-01]
[-6.17553640e-02	1.22543111e-01]
[-6.04744831e-02	1.30066478e-01]
[-5.18077654e-02	8.51264784e-02]
[-5.44148539e-02	1.13166493e-01]
[-6.26032358e-02	1.12291578e-01]
[-6.58281013e-02	1.15026533e-01]
[-5.74631858e-02	1.07484815e-01]
[-6.38743254e-02	1.30866463e-01]
[-5.65149237e-02	1.12376478e-01]
[-6.45090918e-02	1.34978104e-01]
[-6.00419369e-02	1.21854779e-01]
[-9.50422391e-02	-4.11326398e-03]
[-8.94516138e-02	-7.66818721e-03]
[-9.51076111e-02	-1.87065550e-02]
[-7.58192461e-02	-2.59532167e-02]
[-8.91845985e-02	-2.23715412e-02]

[-8.20381882e-02 -2.73115299e-02]
[-9.03091628e-02 -1.61048038e-02]
[-6.72517650e-02 1.26006092e-03]
[-9.00134647e-02 -1.38232718e-02]
[-7.46957758e-02 -1.63848150e-02]
[-6.75193572e-02 -1.74332892e-02]
[-8.31419586e-02 -9.85482711e-03]
[-7.88118832e-02 -1.52266876e-02]
[-8.68093930e-02 -2.77481926e-02]
[-7.68400783e-02 1.00767816e-02]
[-9.06941347e-02 -2.15851353e-05]
[-8.23975875e-02 -2.66247878e-02]
[-7.97614708e-02 -7.02497949e-03]
[-8.39252666e-02 -4.16515405e-02]
[-7.65095933e-02 -1.03366139e-02]
[-8.76667106e-02 -3.34430418e-02]
[-8.24960756e-02 -9.61586108e-04]
[-8.80346931e-02 -4.67765227e-02]
[-8.60633510e-02 -2.69565882e-02]
[-8.68441800e-02 -5.05326495e-03]
[-8.95153150e-02 -4.69991309e-03]
[-9.24270302e-02 -2.36165904e-02]
[-9.40306212e-02 -3.28448113e-02]
[-8.51324237e-02 -2.33031619e-02]
[-7.53750063e-02 1.22366693e-02]
[-7.47961661e-02 -1.10249457e-02]
[-7.40865819e-02 -5.09997886e-03]
[-7.90422093e-02 -2.91492876e-03]
[-8.77228682e-02 -5.53347837e-02]
[-8.08321251e-02 -2.98247694e-02]
[-8.72878429e-02 -9.84644677e-03]
[-9.24729338e-02 -1.39265443e-02]
[-8.42195253e-02 -2.91132749e-02]
[-7.99092041e-02 -6.79486183e-03]
[-7.66114230e-02 -1.97965423e-02]
[-7.89709646e-02 -3.07432189e-02]
[-8.66708741e-02 -2.06798593e-02]
[-7.91807283e-02 -9.98326205e-03]
[-6.76384077e-02 -2.18285462e-04]
[-7.92555461e-02 -2.00198695e-02]
[-8.10515660e-02 -7.24989641e-03]
[-8.08304543e-02 -1.22632043e-02]
[-8.52787176e-02 -8.25324653e-03]
[-6.77844702e-02 1.75733973e-02]
[-7.98997583e-02 -1.13515454e-02]
[-9.88338505e-02 -8.53894903e-02]
[-8.66823361e-02 -6.43396776e-02]
[-1.02672920e-01 -7.00946964e-02]
[-9.38862295e-02 -6.81980592e-02]
[-9.76169023e-02 -7.76396157e-02]
[-1.10328828e-01 -9.00247153e-02]
[-7.52879803e-02 -5.70863507e-02]
[-1.05455794e-01 -8.01281242e-02]
[-9.65020152e-02 -8.20914370e-02]
[-1.07601304e-01 -6.57445578e-02]
[-9.43168735e-02 -3.96830270e-02]

```

[-9.24479381e-02 -6.27197251e-02]
[-9.81862967e-02 -5.89346842e-02]
[-8.47477972e-02 -7.00413173e-02]
[-8.79533083e-02 -7.09361942e-02]
[-9.51282875e-02 -5.50679223e-02]
[-9.53131729e-02 -5.79297443e-02]
[-1.14989851e-01 -6.97229940e-02]
[-1.11480981e-01 -1.16578003e-01]
[-8.50328416e-02 -6.48015027e-02]
[-1.01180373e-01 -6.30279529e-02]
[-8.46187241e-02 -5.84163005e-02]
[-1.10679013e-01 -9.66364242e-02]
[-8.93518003e-02 -4.64247606e-02]
[-9.96610459e-02 -5.92796557e-02]
[-1.04257505e-01 -6.05231151e-02]
[-8.84305502e-02 -4.09564181e-02]
[-8.89746034e-02 -4.03897307e-02]
[-9.47978024e-02 -7.54813178e-02]
[-1.02046160e-01 -5.48298557e-02]
[-1.04948198e-01 -7.55614491e-02]
[-1.14601538e-01 -5.06830826e-02]
[-9.49727792e-02 -7.74162886e-02]
[-9.02921735e-02 -4.55215034e-02]
[-9.04325947e-02 -7.28931692e-02]
[-1.08788476e-01 -7.23446854e-02]
[-9.69165324e-02 -6.44161979e-02]
[-9.49265302e-02 -5.64513979e-02]
[-8.76572648e-02 -3.79997254e-02]
[-9.88305089e-02 -5.02663601e-02]
[-9.88591917e-02 -6.72512462e-02]
[-9.75766400e-02 -4.21663133e-02]
[-8.66823361e-02 -6.43396776e-02]
[-1.01466857e-01 -7.26079359e-02]
[-1.00360953e-01 -6.70195388e-02]
[-9.61496966e-02 -5.24346281e-02]
[-8.92692075e-02 -5.85064019e-02]
[-9.40593040e-02 -4.98296974e-02]
[-9.48896096e-02 -5.61012256e-02]
[-8.84783560e-02 -5.15697045e-02]]

```

```

In [23]: SigmaReduced2 = Sigma[:n_elements, :n_elements]
print("Matriz Sigma Reducida")
print(SigmaReduced2)

```

```

Matriz Sigma Reducida
[[95.95991387  0.
  [ 0.          17.76103366]]

```

```

In [25]: VTReduced2 = VT[:n_elements, :]
print("VT Reducida")
print(VTReduced2)

```

```

VT Reducida
[[-0.75110816 -0.38008617 -0.51300886 -0.16790754]
 [ 0.2841749  0.5467445  -0.70866455 -0.34367081]]

```

```
In [27]: AReduced2 = UReduced2.dot(SigmaReduced2.dot(VTReduced2))  
print("Matriz A transformada:")  
print(AReducd2)
```

Matriz A transformada:

```
[[5.0952927 3.50597743 1.40192232 0.20165319]
 [4.74588049 3.19610853 1.46136967 0.25800276]
 [4.68667405 3.21586325 1.30954904 0.19452725]
 [4.61488457 3.08894388 1.46347879 0.27002699]
 [5.07488651 3.50623125 1.36428119 0.1863997 ]
 [5.52598407 3.7330351 1.67566825 0.28872322]
 [4.731593 3.2288014 1.36216771 0.21446447]
 [5.00510918 3.39830515 1.47931372 0.24418439]
 [4.37933538 2.93134058 1.38864652 0.25618379]
 [4.80551481 3.23360903 1.48569239 0.26393296]
 [5.39533378 3.70766642 1.49514864 0.2183418 ]
 [4.89451945 3.29088668 1.51906398 0.27146211]
 [4.67854319 3.16443092 1.41000708 0.24052709]
 [4.30090163 3.00174374 1.08842179 0.12739443]
 [5.73037625 4.07476895 1.2813685 0.09359359]
 [5.90310008 4.12548386 1.48153824 0.16928199]
 [5.45240789 3.81002345 1.36951483 0.15684908]
 [5.09813811 3.49356935 1.43489086 0.21628344]
 [5.67252863 3.83336074 1.71712893 0.29503728]
 [5.24878923 3.60734078 1.45368795 0.21202774]
 [5.29986481 3.53615373 1.70596661 0.32150311]
 [5.20754895 3.55059319 1.50590357 0.23906608]
 [4.74334279 3.40687684 0.98455159 0.04314391]
 [5.07083969 3.32233292 1.76893717 0.36926573]
 [4.94970159 3.23314542 1.74867904 0.37036772]
 [4.84716047 3.20170004 1.63284042 0.32678525]
 [5.02919404 3.3542419 1.62178914 0.30641344]
 [5.17817863 3.53081603 1.49685472 0.23746713]
 [5.1156989 3.50572362 1.43956345 0.21690668]
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 [4.4931984 3.08360618 1.25436691 0.18599101]
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 [5.32521082 3.51794435 1.7928099 0.35853215]
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[5.68235564 2.85460324 3.92780157 1.29135169]
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[5.67554567 2.65857223 4.35461986 1.4844478]
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[6.1425375 2.96033305 4.526948 1.52270396]
[5.65666399 2.79101665 4.02358701 1.33672831]
[4.87402271 2.46485839 3.33247035 1.09114899]
[5.61139942 2.69628252 4.15359708 1.39919674]
[5.80530296 2.8857948 4.08128132 1.35018765]
[5.76406268 2.82904722 4.13349695 1.37722599]
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[4.97434924 2.64295627 3.1157237 0.98490288]
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[5.8041613 2.51903586 4.90089823 1.71998021]
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[6.88399826 3.05929118 5.65227267 1.96761789]
[7.2090179 3.2148631 5.89419944 2.04926786]
[6.16702774 2.82761324 4.86878708 1.67482363]
[6.20910128 2.85295947 4.88843721 1.68013059]
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[7.07836761 3.18949442 5.71367983 1.97888643]
[7.18289837 3.09402129 6.11748277 2.15218979]
[8.00424223 3.68769544 6.27956573 2.15587192]
[6.45454552 2.7121756 5.64975615 2.00278377]
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[6.15013521 2.59050415 5.36931989 1.90202165]
[7.47592704 3.26532505 6.26604488 2.19443089]
[6.66025859 2.90931078 5.58181665 1.9547504 ]
[6.5570269 2.91407352 5.38360238 1.87406997]
[6.12621535 2.82812087 4.79350481 1.64431665]
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[6.78596906 2.95263502 5.7131341 2.00335628]
[6.82013719 3.14944992 5.33426189 1.82957415]
[5.92300206 2.53678163 5.07704158 1.78938593]
[6.94688336 2.99572703 5.90892758 2.07806896]
[6.8953799 3.00965886 5.78414682 2.02613891]
[6.66546178 2.99769195 5.39325923 1.86925995]
[6.13889606 2.68777816 5.13096763 1.7954606 ]
[6.52794178 2.94674481 5.25756553 1.81967836]
[6.5561332 2.91612734 5.37737736 1.87133766]
[6.1169056 2.72629392 5.00472583 1.74037685]]

```

```

In [29]: T2 = UReduced2.dot(SigmaReduced2)
print("Matriz T (2 Dimensiones):")
print(T2)

```

Matriz T (2 Dimensiones):

```
[[ -5.91274714e+00  2.30203322e+00]
 [ -5.57248242e+00  1.97182599e+00]
 [ -5.44697714e+00  2.09520636e+00]
 [ -5.43645948e+00  1.87038151e+00]
 [ -5.87564494e+00  2.32829018e+00]
 [ -6.47759822e+00  2.32464996e+00]
 [ -5.51597520e+00  2.07090423e+00]
 [ -5.85092859e+00  2.14807482e+00]
 [ -5.15891972e+00  1.77506408e+00]
 [ -5.64500117e+00  1.99000106e+00]
 [ -6.26539771e+00  2.42576813e+00]
 [ -5.75200785e+00  2.02037338e+00]
 [ -5.48058085e+00  1.97777558e+00]
 [ -4.95112411e+00  2.04828749e+00]
 [ -6.52596417e+00  2.91606081e+00]
 [ -6.79037199e+00  2.82500759e+00]
 [ -6.27239468e+00  2.60811578e+00]
 [ -5.92953789e+00  2.26766614e+00]
 [ -6.64813130e+00  2.38959506e+00]
 [ -6.09486463e+00  2.36082303e+00]
 [ -6.25397363e+00  2.12001187e+00]
 [ -6.07364677e+00  2.27178150e+00]
 [ -5.36999813e+00  2.49808604e+00]
 [ -6.04100482e+00  1.87698371e+00]
 [ -5.90591050e+00  1.80777402e+00]
 [ -5.75019501e+00  1.85851056e+00]
 [ -5.93581099e+00  2.00847420e+00]
 [ -6.03915884e+00  2.25958425e+00]
 [ -5.94984934e+00  2.27577626e+00]
 [ -5.60087980e+00  1.88260699e+00]
 [ -5.63798200e+00  1.85635003e+00]
 [ -6.18495337e+00  2.19301062e+00]
 [ -6.25041979e+00  2.62199803e+00]
 [ -6.47925073e+00  2.79842433e+00]
 [ -5.66179193e+00  1.95563398e+00]
 [ -5.62100870e+00  2.25132529e+00]
 [ -6.16188952e+00  2.48656963e+00]
 [ -5.78374337e+00  2.33423977e+00]
 [ -5.14562746e+00  1.90060499e+00]
 [ -5.92603941e+00  2.17649231e+00]
 [ -5.80312619e+00  2.31011510e+00]
 [ -4.97146870e+00  1.51193425e+00]
 [ -5.22164469e+00  2.00995389e+00]
 [ -6.00740111e+00  1.99441449e+00]
 [ -6.31685893e+00  2.04299013e+00]
 [ -5.51416236e+00  1.90904141e+00]
 [ -6.12937476e+00  2.32432366e+00]
 [ -5.42316721e+00  1.99592242e+00]
 [ -6.19028689e+00  2.39735064e+00]
 [ -5.76161909e+00  2.16426683e+00]
 [ -9.12024508e+00 -7.30558200e-02]
 [ -8.58376916e+00 -1.36194931e-01]
 [ -9.12651817e+00 -3.32247752e-01]
 [ -7.27560832e+00 -4.60955955e-01]
 [ -8.55814639e+00 -3.97341697e-01]
```

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[-7.97829519e+00 -1.75031916e-01]
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[-8.33022188e+00 -4.92836582e-01]
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[-7.90686540e+00 -4.72883753e-01]
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[-8.86928986e+00 -4.19455056e-01]
[-9.02317031e+00 -5.83357800e-01]
[-8.16930004e+00 -4.13888242e-01]
[-7.23297912e+00 2.17335895e-01]
[-7.17743366e+00 -1.95814432e-01]
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[-7.58488360e+00 -5.17721477e-02]
[-8.41787888e+00 -9.82802956e-01]
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[-8.87369476e+00 -2.47349822e-01]
[-8.08169840e+00 -5.17081855e-01]
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[-7.35162556e+00 -3.51607055e-01]
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[-6.49057578e+00 -3.87697543e-03]
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[-8.18333840e+00 -1.46586189e-01]
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[-7.66717393e+00 -2.01615180e-01]
[-9.48408779e+00 -1.51660561e+00]
[-8.31802951e+00 -1.14273918e+00]
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```

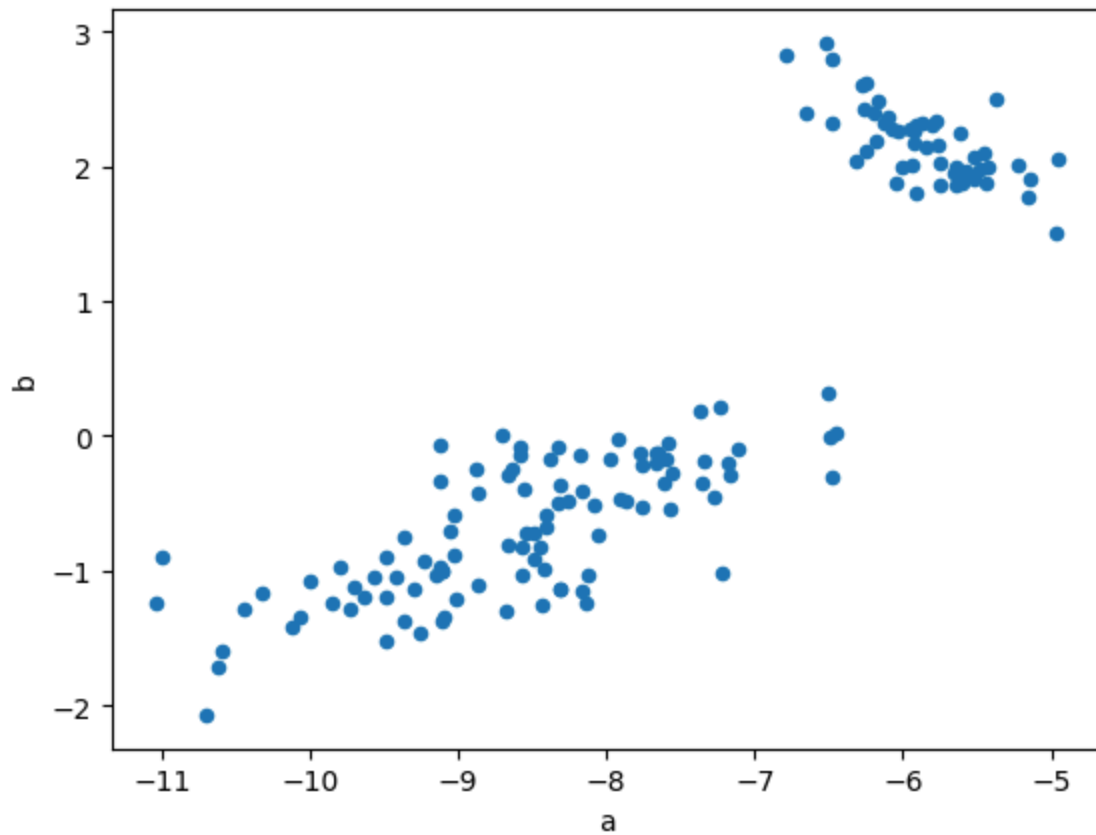
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[-9.10559876e+00 -9.96415757e-01]
[-8.49037542e+00 -9.15931258e-01]]

```

```

In [31]: df = pd.DataFrame(T2, columns=["a", "b"])
df.plot(kind="scatter", x="a", y="b")
plt.show()

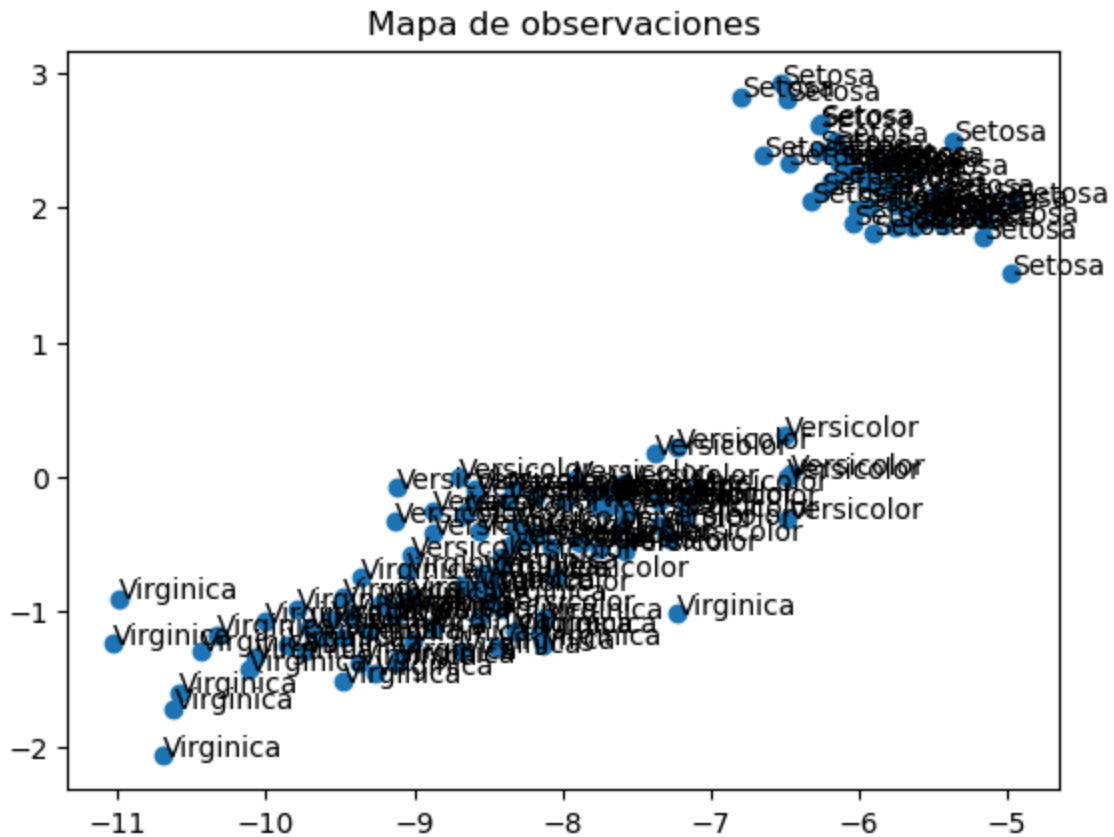
```



```
In [33]: df = pd.DataFrame(T2, columns=["a", "b"])
x = df.iloc[:, 0]
y = df.iloc[:, 1]
x = x.to_numpy()
y = y.to_numpy()

fig, ax = plt.subplots()
ax.set_title("Mapa de observaciones")
ax.scatter(x, y)

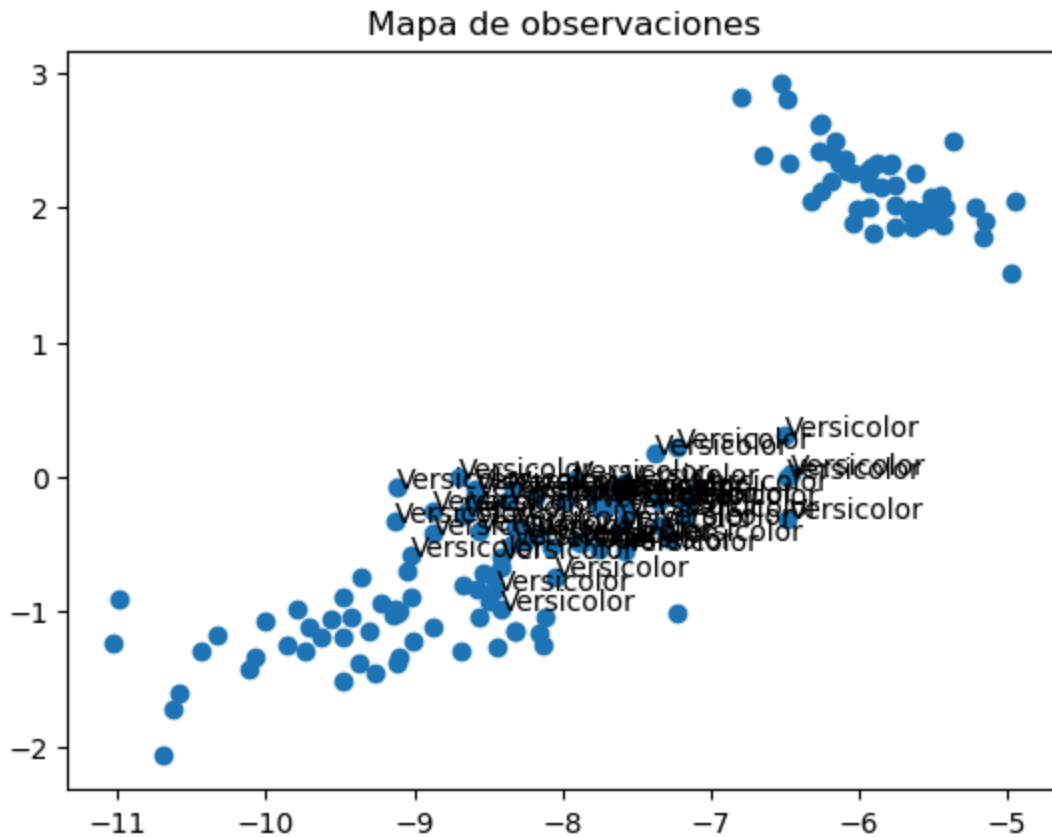
for i, txt in enumerate(etiquetas):
    ax.annotate(txt, (x[i], y[i]))
```



```
In [35]: df = pd.DataFrame(T2, columns=["a", "b"])
x = df.iloc[:, 0]
y = df.iloc[:, 1]
x = x.to_numpy()
y = y.to_numpy()

fig, ax = plt.subplots()
ax.set_title("Mapa de observaciones")
ax.scatter(x, y)

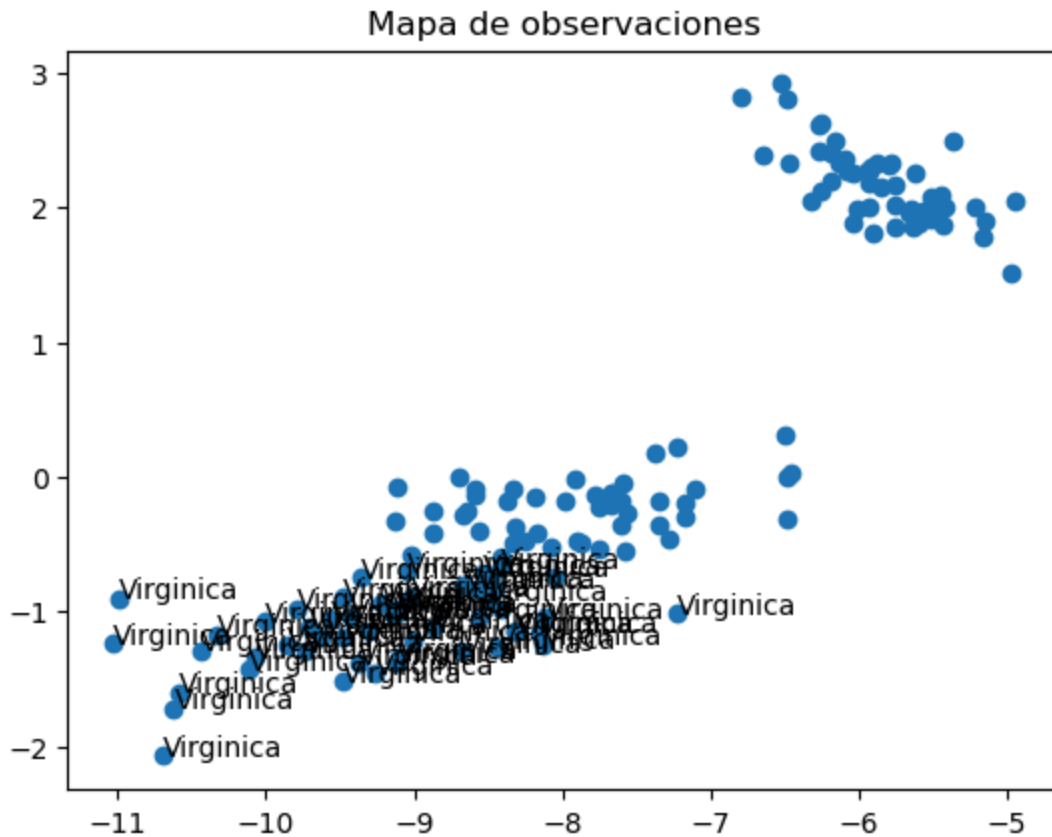
for i, txt in enumerate(etiquetas):
    if txt == "Versicolor":
        ax.annotate(txt, (x[i], y[i]))
```



```
In [37]: df = pd.DataFrame(T2, columns=["a", "b"])
x = df.iloc[:, 0]
y = df.iloc[:, 1]
x = x.to_numpy()
y = y.to_numpy()

fig, ax = plt.subplots()
ax.set_title("Mapa de observaciones")
ax.scatter(x, y)

for i, txt in enumerate(etiquetas):
    if txt == "Virginica":
        ax.annotate(txt, (x[i], y[i]))
```



```
In [39]: # Reducción a 3 Dimensiones
n_elements = 3

UReduced3 = U[:, :n_elements]

SigmaReduced3 = Sigma[:, :n_elements]

VTReduced3 = VT[:, :n_elements]

AReduced3 = UReduced3.dot(SigmaReduced3.dot(VTReduced3))
print("Matriz A transformada:")
print(AReduced3)
```

Matriz A transformada:

```
[[5.09900942 3.50097959 1.4014844 0.19767844]
 [4.86870364 3.03094925 1.44689807 0.12665275]
 [4.69422107 3.20571484 1.30865982 0.18645629]
 [4.62518119 3.0750981 1.4622656 0.25901553]
 [5.01947963 3.58073643 1.37080948 0.24565314]
 [5.40687172 3.89320449 1.68970262 0.41610481]
 [4.61617117 3.38400818 1.37576725 0.33789933]
 [5.01454656 3.38561478 1.47820176 0.23409183]
 [4.40998651 2.89012425 1.38503506 0.22340474]
 [4.91842577 3.08177859 1.4723887 0.14318329]
 [5.40391192 3.69613146 1.49413792 0.20916813]
 [4.8505539 3.3500068 1.5242442 0.31847992]
 [4.80311694 2.99691763 1.39532922 0.10730496]
 [4.30830887 2.99178329 1.08754903 0.11947296]
 [5.74700833 4.05240393 1.27940883 0.07580684]
 [5.69603935 4.40391672 1.5059351 0.39071768]
 [5.34517978 3.95421221 1.38214894 0.27152138]
 [5.07488831 3.52483318 1.43763026 0.24114734]
 [5.68993828 3.80995013 1.71507765 0.27641899]
 [5.12084537 3.77938583 1.46876289 0.34885395]
 [5.40422382 3.39582303 1.69367054 0.20989909]
 [5.08654623 3.71330455 1.52016068 0.36846927]
 [4.59895639 3.60103203 1.00156386 0.19755416]
 [5.05255898 3.34691484 1.77109109 0.38881555]
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In [41]: T3 = UReduced3.dot(SigmaReduced3)
print("Matriz T (3 Dimensiones):")
print(T3)
```

Matriz T (3 Dimensiones):

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```

```

In [43]: from mpl_toolkits import mplot3d
          %matplotlib inline

```

```

In [45]: df3 = pd.DataFrame(T3, columns = ["a", "b", "c"])
          df3

```

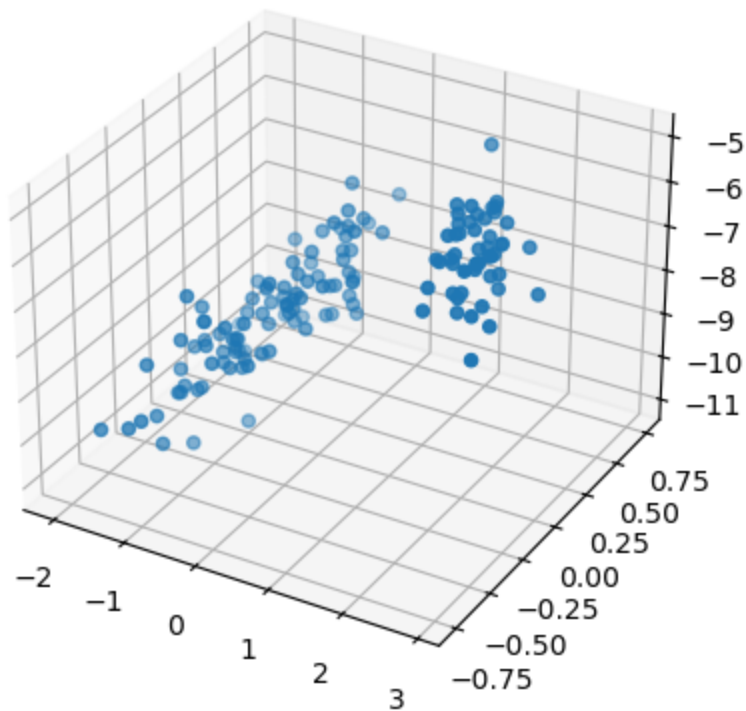
Out[45]:

	a	b	c
0	-5.912747	2.302033	0.007402
1	-5.572482	1.971826	0.244592
2	-5.446977	2.095206	0.015029
3	-5.436459	1.870382	0.020505
4	-5.875645	2.328290	-0.110338
...	...	...	...
145	-9.226517	-0.931293	-0.204095
146	-8.566265	-1.039134	0.159305
147	-9.025923	-0.885027	-0.143421
148	-9.105599	-0.996416	-0.737103
149	-8.490375	-0.915931	-0.331394

150 rows × 3 columns

```
In [47]: fig = plt.figure()
ax = plt.axes(projection="3d")
xline = df3["b"]
yline = df3["c"]
zline = df3["a"]
ax.scatter3D(xline, yline, zline)
```

Out[47]: <mpl_toolkits.mplot3d.art3d.Path3DCollection at 0x1f113052650>



```
In [49]: from mpl_toolkits.mplot3d import Axes3D

fig = plt.figure()
ax3d = fig.add_subplot(projection="3d")

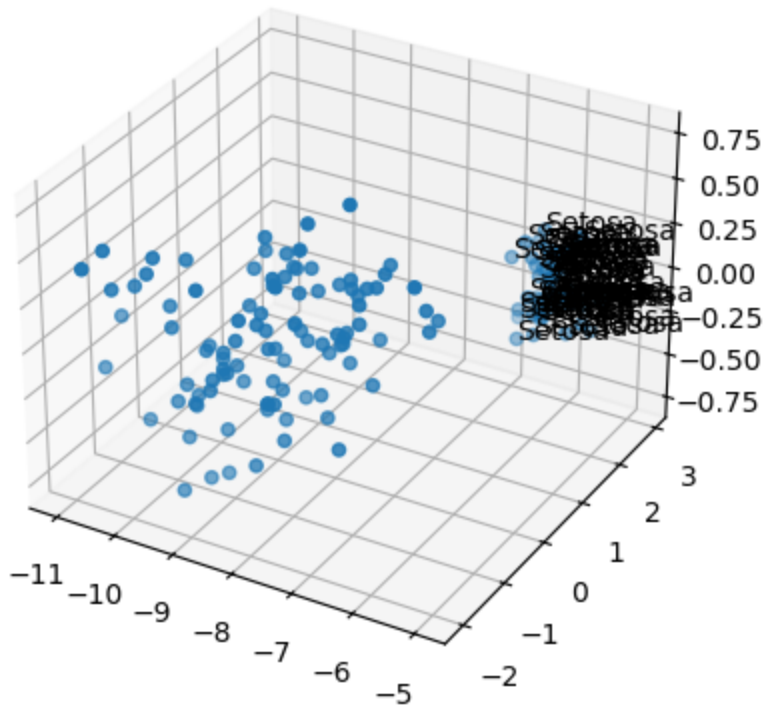
arrayx = np.array(df3["a"])
arrayy = np.array(df3["b"])
arrayz = np.array(df3["c"])

ax3d.scatter(arrayx, arrayy, arrayz)

for x, y, z, label in zip(arrayx, arrayy, arrayz, etiquetas):
    if label == "Setosa":
        ax3d.text(x, y, z, label)

plt.title("Datos en 3D")
plt.show()
```

Datos en 3D



```
In [51]: from mpl_toolkits.mplot3d import Axes3D

fig = plt.figure()
ax3d = fig.add_subplot(projection="3d")

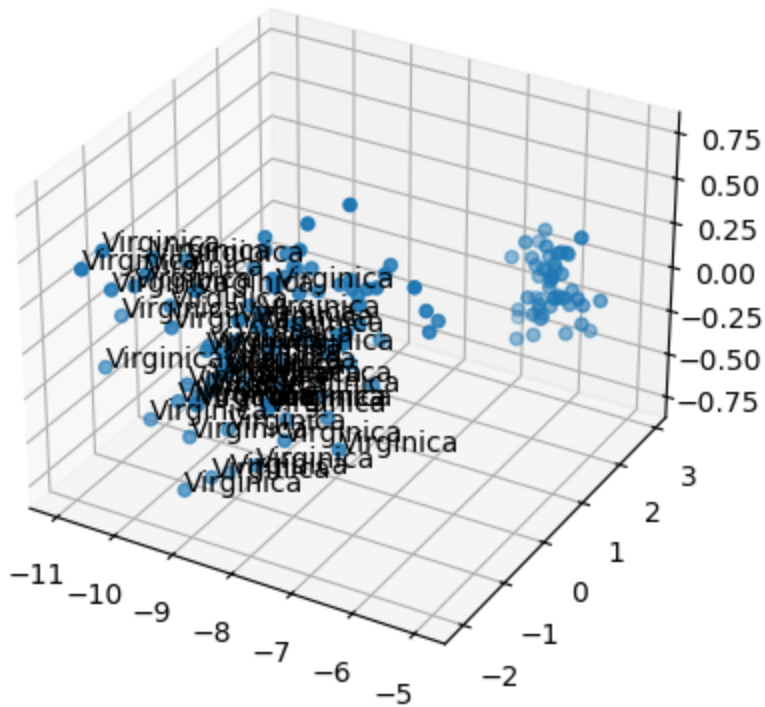
arrayx = np.array(df3["a"])
arrayy = np.array(df3["b"])
arrayz = np.array(df3["c"])

ax3d.scatter(arrayx, arrayy, arrayz)

for x, y, z, label in zip(arrayx, arrayy, arrayz, etiquetas):
    if label == "Virginica":
        ax3d.text(x, y, z, label)

plt.title("Datos en 3D")
plt.show()
```

Datos en 3D



```
In [53]: from mpl_toolkits.mplot3d import Axes3D

fig = plt.figure()
ax3d = fig.add_subplot(projection="3d")

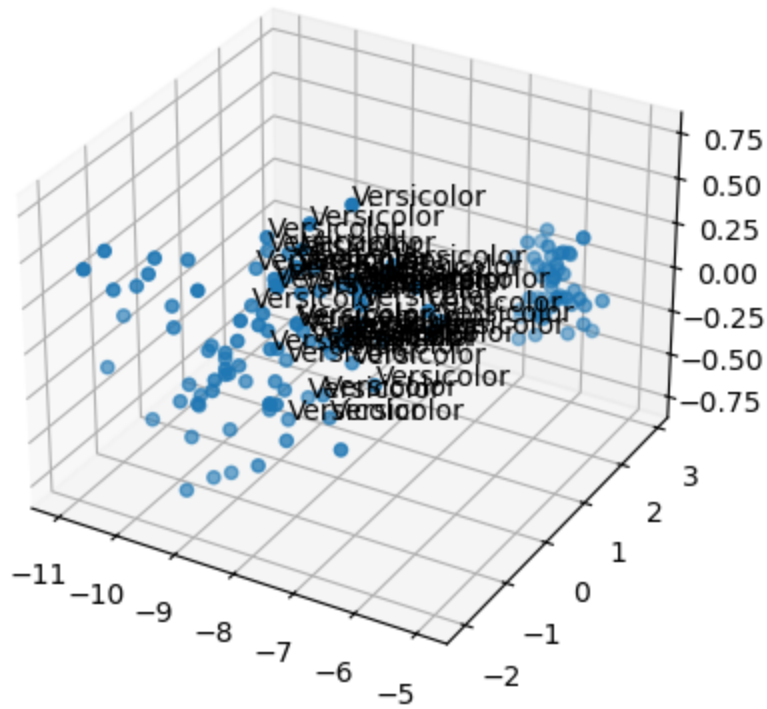
arrayx = np.array(df3["a"])
arrayy = np.array(df3["b"])
arrayz = np.array(df3["c"])

ax3d.scatter(arrayx, arrayy, arrayz)

for x, y, z, label in zip(arrayx, arrayy, arrayz, etiquetas):
    if label == "Versicolor":
        ax3d.text(x, y, z, label)

plt.title("Datos en 3D")
plt.show()
```

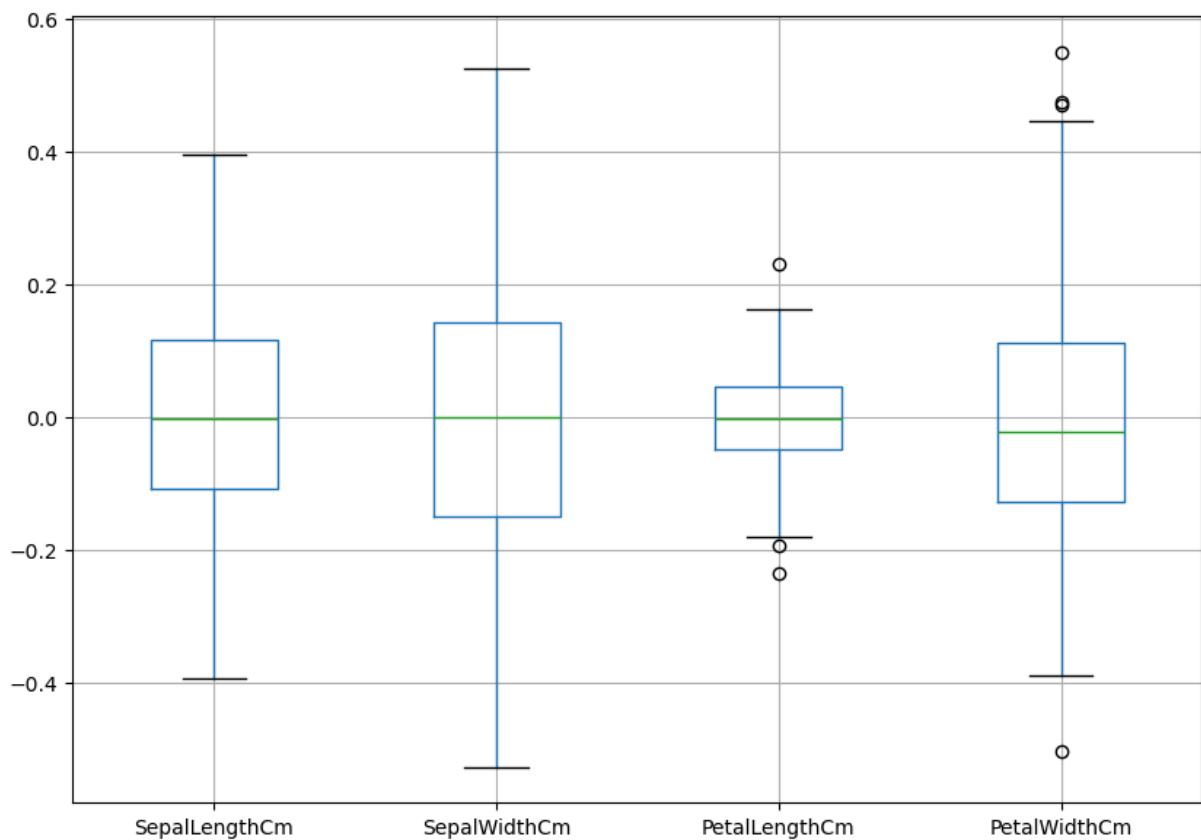
Datos en 3D



```
In [55]: plt.figure(figsize = (10,7))
Error2 = B - AReduced2

dfError2 = pd.DataFrame(Error2, columns = ["SepalLengthCm", "SepalWidthCm", "PetalL
dfError2.boxplot()
```

Out[55]: <Axes: >



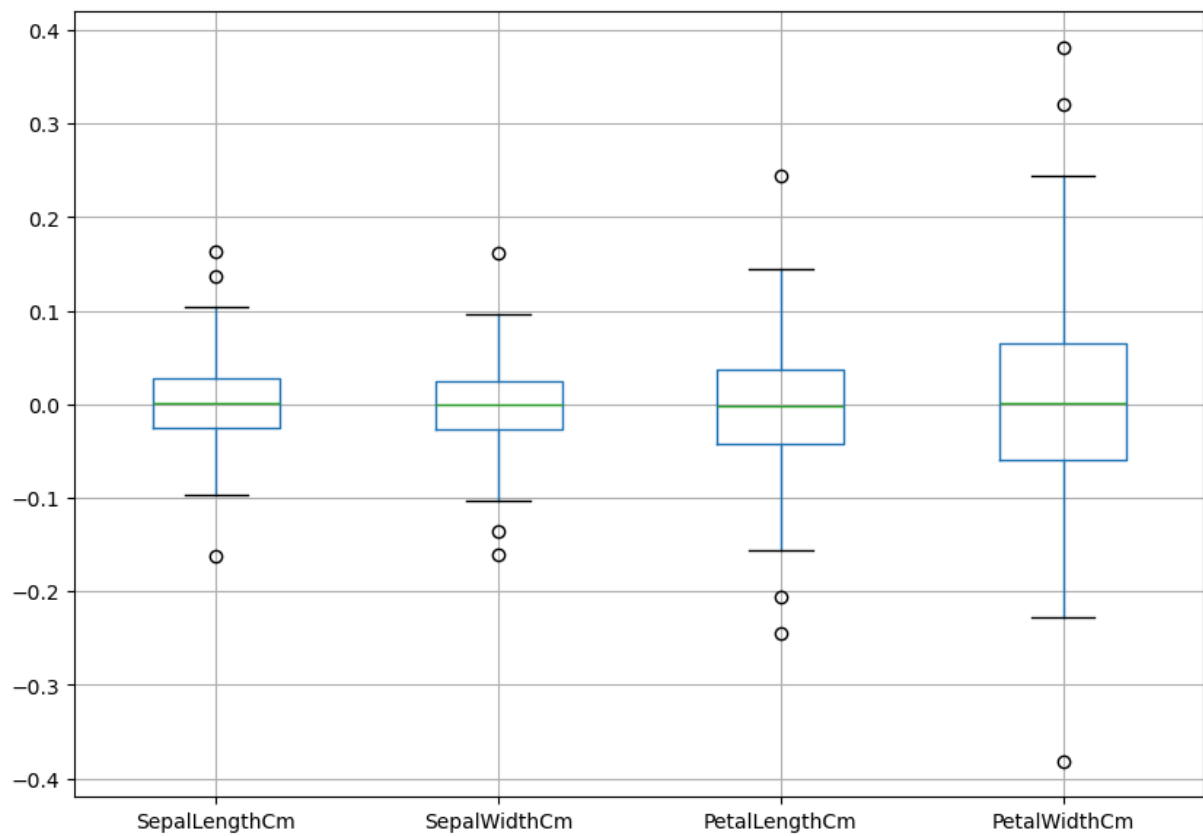
```
In [57]: print("Errores promedio por columna")
         Error2.mean(axis=0)
```

Errores promedio por columna

```
Out[57]: array([ 1.63686453e-03, -2.22711818e-03, -9.18095159e-05, -2.00030563e-03])
```

```
In [59]: Error3 = B - AReduced3

plt.figure(figsize = (10, 7))
dfError3 = pd.DataFrame(Error3, columns = ["SepalLengthCm", "SepalWidthCm", "PetalL
dfError3.boxplot()
plt.show()
```



```
In [61]: print("Errores promedio por columna")  
Error3.mean(axis=0)
```

Errores promedio por columna

```
Out[61]: array([-7.31905331e-05,  7.23787682e-05,  1.09677133e-04, -1.71531926e-04])
```